

Two way ANOVA

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LSM

Consider a drug development trial. There are 'a' different drugs to treat an ailment. Each of these drugs will be applied to 'b' different strata of the population.

Select n_j people at random from j^{th} stratum for $j=1, \dots, b$

Altogether $\sum_{j=1}^b n_j$ subjects

i^{th} drug being applied to n_{ij} subjects corr. to j^{th} stratum so that $\sum_{j=1}^b n_{ij} = n_i$ (Randomized Block Design)

Let Y_{ijk} denote the response (health outcome) corr. to the i^{th} drug being administered to the k^{th} individual within the subgroup of the j^{th} stratum receiving drug i .

$(Y_{ijk})_{\substack{k=1, \dots, n_{ij} \\ j=1, \dots, b \\ i=1, \dots, a}}$

Q1) Is there any effect of the drugs in treating the ailment?

Q2) Is there any effect of the strata?

Q3) Is there any interaction between drugs and strata?

Let drug refers to Factor A

stratum refers to Factor B

There are a levels of Factor A
b levels of Factor B

Further assume that $m_{ij} \geq 1 \quad \forall i, j$
(Complete Design)

If m_{ij} 's are all equal to say m , then we say that the experimental design is balanced.

The combination of i^{th} level of Factor A and j^{th} level of Factor B is referred to as treatment (i, j)

$$Y_{ijk} = \mu_{ij} + \epsilon_{ijk}$$

where ϵ_{ijk} are indep. with zero mean.

If the experimental design is prepared carefully, we hope that the only variation across the treatment groups will be their means.

$$\Rightarrow \text{Var}(\epsilon_{ijk}) = \sigma^2 \text{ (Same)}$$

Recall one-factor (as one-way) ANOVA

$$Y_{ik} = \mu_i + \epsilon_{ik} \quad \begin{array}{l} h=1, \dots, m_i \\ i=1, \dots, a \end{array}$$

$$\boxed{\mu_i = \mu_0 + \alpha_i}$$

ϵ_{ik} are indep. with zero mean and variance σ^2

α_i = treatment effect of the i^{th} treatment

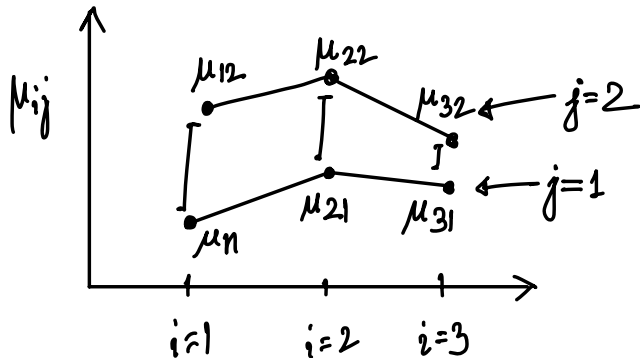
μ_0 = overall mean (of effect)

Modelling μ_{ij}

$$\mu_{ij} = \mu_{..} + \alpha_i + \beta_j + \gamma_{ij}$$

Graphical representation (interaction plot)

For simplicity, suppose $b=2$ and $a=3$



Under appropriate identifiability constraints on the parameters $(\alpha_i), (\beta_j), (\gamma_{ij})$, we can show that
no interaction $\equiv (\gamma_{ij}=0 \forall i, j)$

For now, we restrict ourselves to the balanced design i.e. when $m_{ij}=m \forall i, j$

In this case, the natural identifiability constraints are

$$\begin{aligned} \sum_{i=1}^a \alpha_i &= 0 & \sum_{j=1}^b \beta_j &= 0 & \text{for all } j=1, \dots, b \\ \sum_{i=1}^a \gamma_{ij} &= 0 & \sum_{j=1}^b \gamma_{ij} &= 0 & \text{for all } i=1, \dots, a \end{aligned}$$

Claim: Since $\sum_{i=1}^a \sum_{j=1}^b \gamma_{ij} = \sum_{j=1}^b \sum_{i=1}^a \gamma_{ij} = 0$, there are in effect $(a+b+1) = (1 + (a-1) + (b-1) + 1 + 1)$ linearly independent

constraints on the parameters

Q3) No interaction $\longleftrightarrow \gamma_{ij} = 0 \quad \forall i, j$

Q1) main effect of Factor A (drug) $\longleftrightarrow$ whether α_i 's are all zero

Q2) main effect of Factor B (strata) $\longleftrightarrow$ whether β_j 's are all zero

Notice that under these identifiability constraints

$$\mu_{..} = \left(\sum_{i=1}^a \sum_{j=1}^b \mu_{ij} \right) / ab = E(\overline{Y_{...}})$$

$$\mu_{..} + \alpha_i = \left(\sum_{j=1}^b \mu_{ij} \right) / b = E(\overline{Y_{i..}})$$

$$\mu_{..} + \beta_j = \left(\sum_{i=1}^a \mu_{ij} \right) / a = E(\overline{Y_{.j.}})$$

$$\gamma_{ij} = \mu_{ij} - (\mu_{..} + \alpha_i + \beta_j)$$

$$= \mu_{ij} - (\mu_{..} + \alpha_i) - (\mu_{..} + \beta_j) + \mu_{..} = E(\overline{Y_{ij.}})$$